

# KUSHAGRA SINGHAL

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## SUMMARY

AI Engineer building 5 AI-powered products across education, defense, and fleet intelligence. Specializing in multi-agent systems, RAG pipelines, computer vision (YOLOv10), and LLM orchestration with LangChain and LangGraph.

## WORK EXPERIENCE

### Founding Engineer – Datavedam, Bangalore

Dec 2025 – Present

- Spearhead 5 AI products at a pre-revenue startup, owning ML pipelines and DevOps.
- Built 4-agent Virtual Principal with RAG pipeline using LangChain and pgvector.
- Trained YOLOv10 model achieving 95% mAP for counter-drone geofencing and alerts.
- Designed 11 AI agents for fleet intelligence with V2V and drowsiness detection.
- Automated QA with Robot Framework and pytest via GitHub Actions across 5 products.

### Data Analyst Intern – Nineleaps, Bangalore

Mar 2025 – Aug 2025

- Introduced LLM agent (LangGraph) for Git issue triage and metric suggestions.
- Delivered 4 analytics modules and BI dashboards for 30+ stakeholders.
- Optimized SQL pipelines processing 1M+ rows, improving performance by 35%.

## PROJECTS

### Lakshya Vedh – Sports AI Coach

- Created pose estimation engine (MediaPipe) analyzing biomechanics of 50+ athletes.
- Integrated Ollama (Llama 3.1 8B) for bilingual coaching in rural health camps.
- Presented automated scoring system to Bihar Health Minister for state sports program.
- **Tech:** MediaPipe, Ollama, Gradio, Python, FastAPI

### BroAI – Converged Cognitive Architecture

- Engineered unified PostgreSQL memory store replacing 3 databases, cutting costs by 60%.
- Achieved 92% retrieval accuracy with hybrid search (pgvector + BM25 + RRF).
- Deployed as grievance portal processing 1,000+ requests with PII masking.
- **Tech:** FastAPI, PostgreSQL, pgvector, LangChain, Redis, Docker

### ShikshaVedam – AI-First School ERP

- Developed RAG pipeline (pgvector + sentence-transformers) powering 4 AI agent workflows.
- Implemented AI-powered OCR grading engine for automated answer sheet evaluation.
- Structured multi-tenant schema with Keycloak RBAC serving 500+ users.
- **Tech:** FastAPI, LangChain, pgvector, Keycloak, Docker

## TECHNICAL SKILLS

- **Languages:** Python, C#, SQL
- **AI/ML:** LangChain, LangGraph, RAG Pipelines, YOLOv10, OpenCV, MediaPipe, Ollama, Vertex AI, Prompt Engineering
- **Backend:** FastAPI, Pydantic, REST APIs, WebSockets, Gradio, Microservices Architecture
- **Databases:** PostgreSQL, pgvector, Redis, BigQuery, SQLAlchemy, Vector Databases
- **DevOps:** Docker, Docker Compose, GitHub Actions, CI/CD Pipelines, Keycloak, Nginx
- **Frontend:** React, Next.js, Tailwind CSS, React Native, Vite
- **Testing:** pytest, Robot Framework, Unit and Integration Testing, Test Automation
- **Agile:** Scrum Master, Cross-functional Collaboration, Sprint Planning, Jira, Confluence

## EDUCATION

### Bachelor of Technology in Computer Science and Engineering

2021 – 2025

Christ University, Bangalore

## CERTIFICATIONS

- **Google Data Analytics Professional Certificate** – Coursera
- **AI Agents with LangChain and LangGraph** – IBM